keerthanakuppan15-jpg / LU-Decomposition

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Public repository · Forked from AkilaMohan/LU-Decomposition

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This branch is 2 commits ahead of `AkilaMohan/LU-Decomposition:main` . | Contribute | Sync fork

keerthanakuppan15-jpg Add code examples for LU decomposition in README

2f7a845 · yesterday

LICENSE	Initial commit	5 years ago
LU.py	final	3 years ago
README.md	Add code examples for LU decom...	yesterday

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LU Decomposition

AIM:

To write a program to find the LU Decomposition of a matrix.

Equipments Required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm

1. Start the program

2. Import the necessary libraries(numpy,scipy.linalg)
3. Define the matrix using numpy
4. Use lu(),lu_solve(),lu_factor() to get the solutions
5. End the program

Program:

(i) To find the L and U matrix

```
/*
Program to find the L and U matrix.
Developed by: keerthana k
RegisterNumber: 212225230137
*/

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu
A=np.array(eval(input()))
P, L, U = lu(A)
print(L)
print(U)
```

(ii) To find the LU Decomposition of a matrix

```
/*
Program to find the LU Decomposition of a matrix.
Developed by: keerthana k
RegisterNumber: 212225230137
*/

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"

import numpy as np
from scipy.linalg import lu_factor, lu_solve
A = np.array([[3, 2, 7], [2, 3, 1], [3, 4, 1]])
b = np.array([4, 5, 7])
lu, piv = lu_factor(A)
x=lu_solve((lu, piv),b)
print(x)
```

Output:

```python
'''Developed by: Keerthana K
RegisterNumber: 212225230137'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu
A=np.array(eval(input()))
P, L, U = lu(A)
print(L)
print(U)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]	[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]	✓
✓	[[5, 1, 8], [4, 5, 7], [8, 9, 1]]	[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.]] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]	[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.]] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]	✓

Passed all tests! ✓

```python
'''Program to solve a matrix using LU decomposition.
Developed by: keerthana k
RegisterNumber: 212225230137
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"

import numpy as np
from scipy.linalg import lu_factor, lu_solve
A = np.array([[3, 2, 7], [2, 3, 1], [3, 4, 1]])
b = np.array([4, 5, 7])
lu, piv = lu_factor(A)
x=lu_solve((lu, piv),b)
print(x)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]	[0.875 1.125 -0.125]	✓

Passed all tests! ✓

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Languages

- **Python** 100.0%

Suggested workflows

Based on your tech stack

Python Package using Anaconda

Configure

Create and test a Python package on multiple Python versions using Anaconda for package management.

Python application

Configure

Create and test a Python application.

Django

Configure

Build and Test a Django Project

More workflows

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